

Ex. No. : 03

Date:

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A PYTHON PROGRAM TO IMPLEMENT LOGISTIC MODEL

Aim:

To implement a **Logistic Regression model in Python** to classify **Diabetes dataset**

Requirements:

Jupyter Notebook with Python, diabetes.csv dataset, and required Python libraries.

PROGRAM:

Step 1: Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix,
classification_report
```

Step 2: Load Dataset

```
data = load_diabetes(as_frame=True).frame
data.head()
```

	age	sex	bmi	bp	s1	s2	s3	s4	s5	s6	target
0	0.038076	0.050680	0.061696	0.021872	-0.044223	-0.034821	-0.043401	-0.002592	0.019907	-0.017646	151.0
1	-0.001882	-0.044642	-0.051474	-0.026328	-0.008449	-0.019163	0.074412	-0.039493	-0.068332	-0.092204	75.0
2	0.085299	0.050680	0.044451	-0.005670	-0.045599	-0.034194	-0.032356	-0.002592	0.002861	-0.025930	141.0
3	-0.089063	-0.044642	-0.011595	-0.036656	0.012191	0.024991	-0.036038	0.034309	0.022688	-0.009362	206.0
4	0.005383	-0.044642	-0.036385	0.021872	0.003935	0.015596	0.008142	-0.002592	-0.031988	-0.046641	135.0

Step 3: Dataset Information

```
data.info()
data.describe()
```

```
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data.describe()
```

```
... <class 'pandas.core.frame.DataFrame'>
RangeIndex: 442 entries, 0 to 441
Data columns (total 11 columns):
 #   Column  Non-Null Count  Dtype  
---  --
 0   age     442 non-null    float64
 1   sex     442 non-null    float64
 2   bmi     442 non-null    float64
 3   bp      442 non-null    float64
 4   s1      442 non-null    float64
 5   s2      442 non-null    float64
 6   s3      442 non-null    float64
```

Step 4: Define Features and Target

Target column = Class

0 → Low Risk

1 → High Risk

```
data['Class'] = (data['target'] >
data['target'].mean()).astype(int)
```

```
X = data.drop(["target", "Class"], axis=1)
y = data["Class"]
```

Step 5: Split Training and Testing Data

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

Step 6: Train Logistic Regression Model

```
model = LogisticRegression(max_iter=1000)
```

```
model.fit(X_train, y_train)
```

```
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

```
... LogisticRegression
LogisticRegression(max_iter=1000)
```

Step 7: Predict Results

```
y_pred = model.predict(X_test)
```

Step 8: Model Evaluation

Accuracy

```
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

```
Accuracy: 0.7640449438202247
```

Confusion Matrix

```
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred))
```

```
Confusion Matrix:
[[49  1]
 [20 19]]
```

Classification Report

```
print("Classification Report:")
print(classification_report(y_test, y_pred))
```

```
Classification Report:
              precision    recall  f1-score   support

     0       0.71      0.98      0.82        50
     1       0.95      0.49      0.64        39

 accuracy          0.76          89
 macro avg         0.83          89
weighted avg         0.82          89
```

Result:

Thus, the **Logistic Regression** model was successfully implemented using **Python** and evaluated using the **Diabetes** dataset.